ECE 315

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Lecture 14 :

- Ethernet LAN
- CAN
- USB

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Last Lecture

- Hardware Abstraction Layer – HAL
- RP2040 UART Example

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This Lecture

- More Serial…
- Ethernet LAN
- CAN
- USB

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Learning Objectives

- Understand Ethernet's history, frame structure, data rates, and transmission media.
- Describe how Ethernet implements CSMA/CD (Carrier Sense Multiple Access with Collision Detection).
- Identify where Ethernet fits within the OSI networking model (Layers 1 and 2).
- Describe the Controller Area Network (CAN) protocol
- Explain its architecture, frame structure, and arbitration mechanism.
- Discuss how CAN ensures reliable, deterministic communication between multiple nodes on a shared bus.

Learning Objectives

- Recognize typical CAN applications (automotive, industrial, satellite).
- Describe the Universal Serial Bus (USB) standard
- Understand USB's host-device architecture, topologies, and device identification.
- Explain USB frame structures, data transfer types (control, bulk, interrupt, isochronous), and speed evolution (1.0 → 3.x).
- Compare and contrast these three serial communication standards
- Evaluate differences in physical signaling, data rates, topology, and communication roles (bus, star, tree, mesh).

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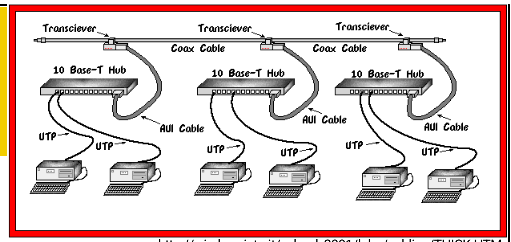
Ethernet – History



http://wireless.ictp.it/school_2001/labo/cabling/THICK.HTM

- Developed at Xerox PARC (1973)
- Inspired by ALOHA network
- Standardized as IEEE 802.3 (1983); used widely
- Initially 10 Mbps over coax (10BASE-5)
- Only one transmitter station can transmit at a time.
- If two (or more) transmitters happen to start transmitting at the same time, then the resulting "***collision***" is detected by those transmitters.
- After a collision, transmitters wait for a random back-off time before trying to transmit again (ALOHA protocol)

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Ethernet Evolution

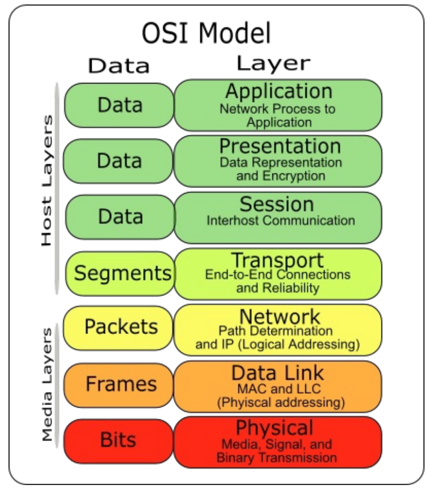
- Ethernet speeds have steadily increased over the years
- Result of improved electronics and signal processing
- Still very much a viable and popular medium
- Faster than wireless
- Spatial multiplexing
  - WiFi channels limited
  - Physical wires can occupy the same space

Standard	Speed	Medium	Notes
10BASE-T	10 Mbps	UTP Cat3	Legacy
100BASE-TX	100 Mbps	UTP Cat5	"Fast Ethernet"
1000BASE-T	1 Gbps	Cat5e/Cat6	Common today
10GBASE-T	10 Gbps	Cat6a	Data centers
40G/100G Ethernet	Multi-lane fiber	Backbone	
2.5G/5GBASE-T	Intermediate	Wi-Fi AP backhaul	

OSI Model – the 7 Layers

- The OSI (Open Systems Interconnection) model defines how network systems communicate.
- Developed by ISO to promote interoperability among network devices.
- Seven (7) layers, each responsible for specific communication functions

1. Physical – Transmission of raw bits (signals, cables)
2. Data Link – Framing, MAC addressing, error detection
3. Network – Routing and IP addressing
4. Transport – Reliable delivery (TCP, UDP)
5. Session – Manages connections between devices
6. Presentation – Data format translation (encryption, compression)
7. Application – User-level protocols (HTTP, SMTP, DNS)



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"File:Osi-model-7-layers.png." *Wikimedia Commons*. 1 Oct 2021, 03:50 UTC. <https://commons.wikimedia.org/w/index.php?title=File:Osi-model-7-layers.png&oldid=594764917> 12 Nov 2025, 02:17.

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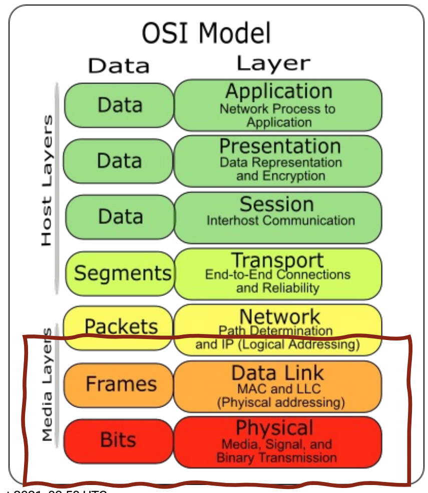
Ethernet operates at Layer 1 (Physical) and Layer 2 (Data Link)

- Provides frame formatting, MAC addressing, and error detection using cyclic redundancy check (CRC)
- Medium access control (MAC) using CSMA/CD in legacy versions
- Defines cabling standards and data rates (e.g., Cat5e, fiber, 1 Gbps, 10 Gbps).



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Ethernet Frame Format

- ***Preamble***: 7 bytes (56 bits) of alternating 0s and 1s for synchronization
- ***Start Frame Delimiter*** (SFD): 1 byte = 0b10101011
- ***Destination Address*** (DA): 6 bytes specifying the destination station
  - An example address, given in hex, is: 07-01-02-01-2C-4B
- ***Source Address*** (SA): 6 bytes specifying the source station
- ***Length / Type***: 2 bytes; called the EtherType in recent standards.
  - If <1518, then this field specifies the length of the data field
  - If >1536 (0x600), then this field specifies the upper layer LAN protocol
- ***Data***: Anywhere from 46 to 1500 bytes of data
- ***Frame Check Sequence*** (FCS): 4 bytes computed using CRC-32

Preamble	SFD	DA	SA	Type	Data	FCS
7 bytes	1 byte	6 bytes	6 bytes	2 bytes	46 to 1500	4 bytes

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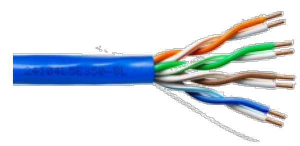
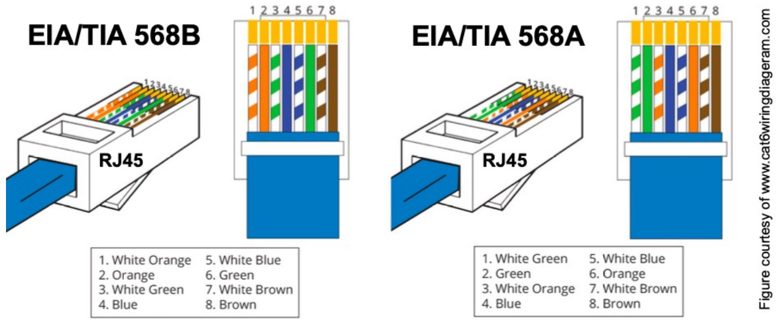
Ethernet Data Rates and Transmission Media

- Ethernet is available for several different data rates (10, 100, 1000, 10000 Mbps, etc.) and a variety of media:
  - **RG-8** thick 50-Ω coaxial cable, for 10-Mbps (10BASE5, the original Ethernet standard). Now obsolete.
  - **RG-58** "thin" 50-Ω coaxial cable, for 10-Mbps (10BASE2).

  Coax cables have been replaced by point-to-point twisted pair links.
  - **CAT-3** twisted pair, for 10-Mbps (10BASE-T)
  - **CAT-5e** twisted pair, for both 100-Mbps (100BASE-TX, a.k.a. Fast Ethernet) and 1-Gbps (1000BASE-T or 1GBASE-T)
  - **CAT-6a**, for 10-Gbps (10GBASE-T)
  - **CAT-7**, intended for 10-Gbps, but CAT-6a is now used instead
  - **CAT-8**, for short range (5 to 30m) at 25 or 40 Gbps
  - **Optical fibre**, for 100-Mbps (100BASE-FX), 1-Gbps (1000BASE-SX),10-Gbps (10GBASE-LX4), 40-Gbps (40GBASE-SR4), etc.



RJ45 Connectors – Type A and B



- Straight ***patch cables*** are A↔A or B↔B
- ***Crossover cables*** are A↔B or B↔A

Straight-through and Crossover Ethernet Cables

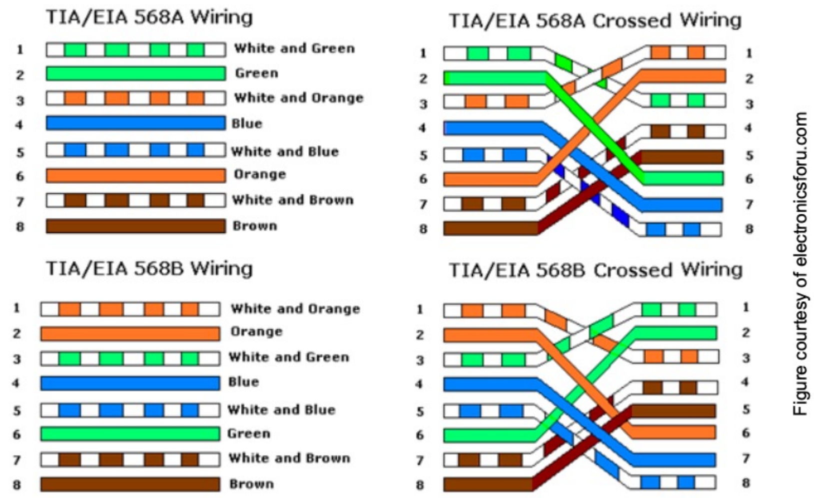


Figure A

Shows the Pin Out of Straight through Cables

Figure B

Shows the Pin Out of Crossover Cables

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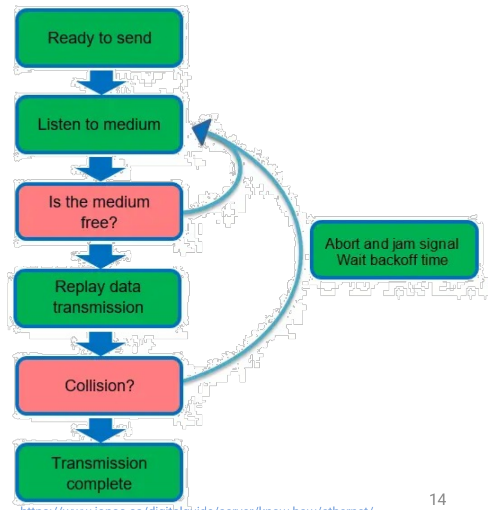
CSMA/CD and Media Access

- ***Carrier Sense Multiple Access with Collision Detection*** (CSMA/CD)
- Nodes listen before transmitting
- If collision → corrupt signals → random backoff (Binary Exponential Backoff)
- Used in half-duplex hubs; not used in full-duplex switched Ethernet



https://www.ionos.ca/digitalguide/server/know-how/ethernet/

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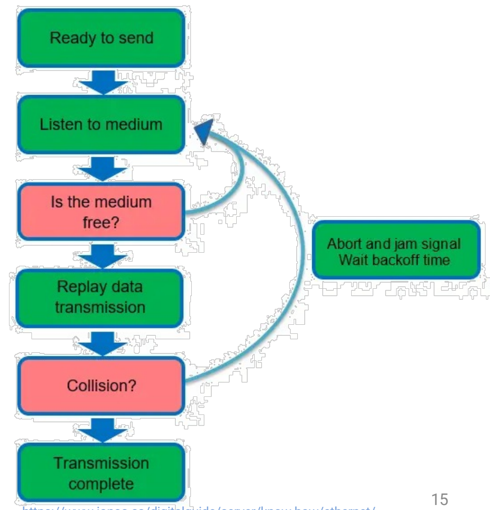

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This is the key to making Ethernet work.

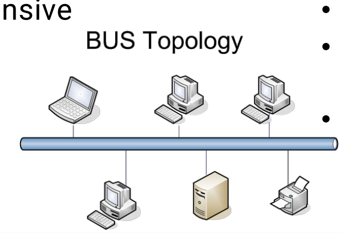
Note, we have Collision Detection, ***not avoidance***

Ethernet Topologies – Bus (Legacy)

- Used in: Early Ethernet (10BASE5 “Thicknet” and 10BASE2 “Thinnet”)
- All nodes connect to a single coaxial cable (the “bus”)
- Each node listens for transmissions and shares the same medium
- Collisions can occur if two nodes transmit simultaneously — handled by CSMA/CD

Advantages:

- Simple and inexpensive for small setups.
- Minimal cabling.



Disadvantages:

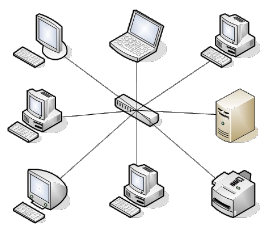
- Collisions reduce performance.
- A single cable fault brings down the entire network.
- Difficult to troubleshoot and scale.

https://whatsincomputer.blogspot.com/2010/08/types-of-network-topologies.html

Ethernet Topologies – Star (Modern)

- Used in: 10BASE-T, 100BASE-TX, 1000BASE-T, etc.
- Each device connects via a dedicated twisted-pair cable to a central switch or hub.
- The central device manages data forwarding between nodes.
- No collisions in full-duplex switched systems.

Advantages:
- High reliability — if one link fails, others remain unaffected.
- Easier troubleshooting and scalability.
- Supports higher data rates.



Disadvantages:
- Requires more cabling.
- Central switch is a single point of failure.

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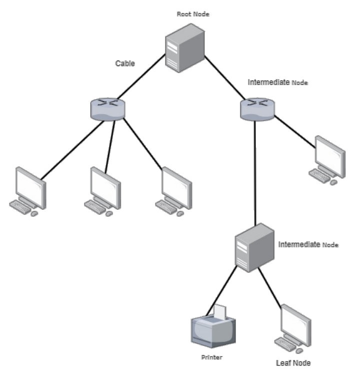
https://en.wikipedia.org/w/index.php?title=Star_network&oldid=1315583823

Ethernet Topologies – Extended Star / Tree

- Used in: Large LANs and enterprise networks
- Combines multiple star networks via interconnected switches or hubs.
- Each branch or layer forms part of a hierarchical tree.

Advantages:
- Scalable for large organizations.
- Structured for easier management.



Disadvantages:
- Switch hierarchy increases cost and complexity.
- Still depends on central backbone stability.

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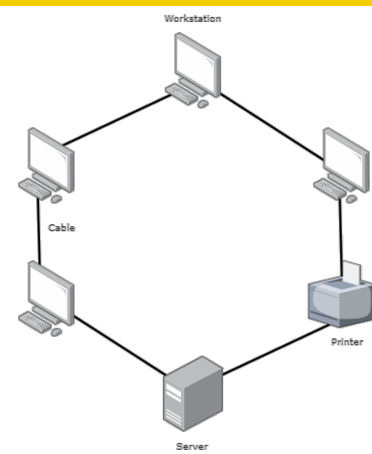
https://www.cybb0rg.com/2024/05/18/network-topologies-explained/

Ethernet Topologies – Ring (rare)

- Used in: Some legacy Ethernet or industrial systems with redundancy (via protocols like Ethernet Ring Protection Switching).
- Nodes connected in a closed loop — each node connected to two others.
- Typically used to provide redundant paths for fault tolerance.
- Managed by spanning tree or rapid ring recovery protocols.



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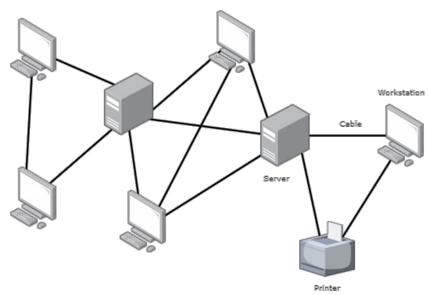
https://www.cybb0rg.com/2024/05/18/network-topologies-explained/

Ethernet Topologies – Mesh (High-Availability Networks)

- Used in: Data centers and industrial control systems requiring high reliability.
- Each node connects to multiple others, providing redundant paths.
- Ensures connectivity even if several links fail.

Advantages:
- Excellent fault tolerance.
- Load balancing possible across multiple paths.



Disadvantages:
- Expensive and complex cabling.
- Requires advanced routing/switching management.

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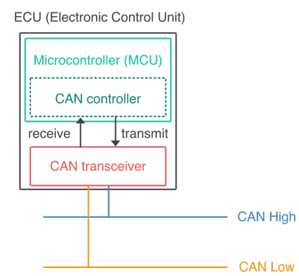
https://www.cybb0rg.com/2024/05/18/network-topologies-explained/

Controller Area Network - CAN

- Mainly used in automotive applications, also within satellite systems
- Developed by Bosch (1983) for automotive electronics
- Replaced point-to-point wiring between Electronic Control Units (ECUs)
- Enables real-time, reliable multi-master comms over a shared bus

Features

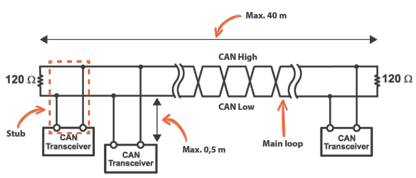
- Multi-master, message-oriented protocol
- Deterministic, priority-based arbitration
- High noise immunity and fault tolerance
- Error detection and confinement
- Speeds up to 1 Mbps (CAN), 5–8 Mbps (CAN FD)

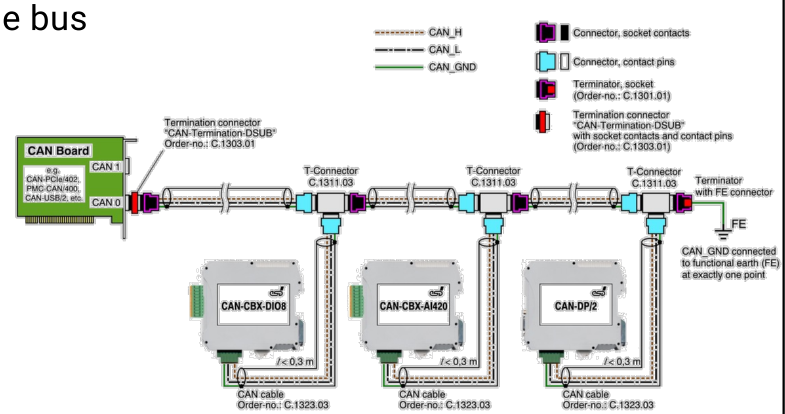


Made with help from ChatGPT https://canlogger1000.csselectronics.com/img/ecu-electronic-control-unit-can-bus-node.svg

CAN Bus Architecture

- Two-wire differential signaling: CAN_H and CAN_L
- All nodes share the same bus
- Each node: MCU + CAN controller + transceiver





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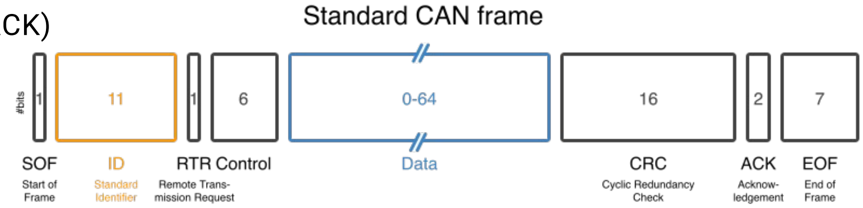
https://esd.eu/fileadmin/_processed_/1/5/csm_can-verdrahtung_beispiel_2-leitungen_en_23_cc85e72226.png

CAN Frame Structure

- Start of Frame (SOF)
- Identifier (ID): 11-bit (standard) or 29-bit (extended)
- Remote Transmission Request (RTR)
- Control: data length
- Data field: 0–8 bytes (up to 64 bytes in CAN FD)
- CRC ensures data integrity
- Acknowledgement (ACK)
- End of Frame (EOF)



https://canlogger1000.csselectronics.com/img/CAN-bus-frame-standard-message-SOF-ID-RTR-Control-Data-CRC-ACK-EOF.svg

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CAN Bus Arbitration

An analogy to keep in mind;

- It's like a group of people politely speaking in turns — each one listens while talking, and if someone more important (lower ID) starts speaking, others stop immediately and try again later, ensuring order without confusion

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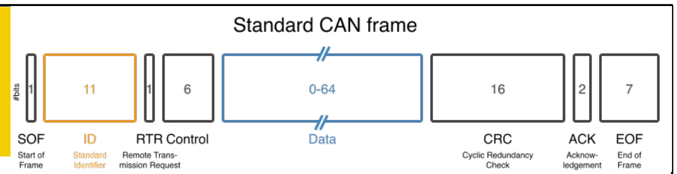
CAN Bus Arbitration



- Every CAN message has an identifier field (11 bits in Classical CAN, 29 bits in Extended CAN).
  - The lower the identifier value, the higher the priority
  - Example: ID `0x100` has higher priority than `0x200`
- CAN uses differential signaling
  - Dominant bit (logical 0): actively drives CAN_H high and CAN_L low
  - Recessive bit (logical 1): bus floats — if any node drives a dominant bit, it overrides recessive
- If one node sends ‘0’ and another sends ‘1’ simultaneously, the bus reads as ‘0’ (dominant).

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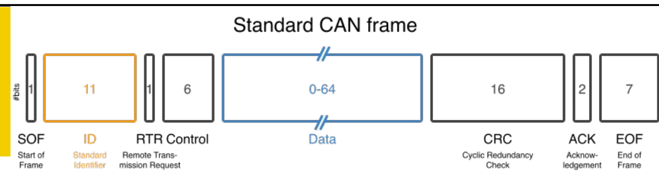
CAN Bus Arbitration



Bitwise Arbitration Process

- All nodes monitor the bus while transmitting
- As soon as a node detects that the bit level on the bus doesn't match what it sent, it knows it lost arbitration and stops transmitting
- The winning node continues uninterrupted
- For example, two nodes A `@0x100` and B `@0x200`

Bit Position	Node A (ID=0x100)	Node B (ID=0x200)	Bus Result	Winner
1	0	0	0	both
2	0	1	0	A continues, B loses

Universal Serial Bus – USB

- Developed in the mid-1990s by Intel, Microsoft, IBM, and others
- Replaced multiple legacy ports (serial, parallel, PS/2)
- Goal: Plug and Play, Hot Swappable, Universal Connector
- USB 1.0 released in January 1996
    - 4 wires (ground, power, differential data pair)
    - Only the 1.5 Mbits/s rate was supported (e.g., keyboard, mouse)
- USB 1.1 released in September 1998
    - Updated standard corrected several flaws in USB 1.0.

Universal Serial Bus – USB

- USB 2.0 released in April 2000
  - Medium (12 Mbit/s) and high speed (480 Mbits/s) added.
  - Backward compatible with USB 1.1.
  - Medium and high speed operation requires shielded cable.
- USB 3.0 released in November 2008
  - SuperSpeed mode (3.2 Gbit/s) added.
  - Adds a shield and two more differential data pairs.
  - Data signaling uses advanced techniques similar to PCIe.
  - New power modes added (idle, sleep, suspend).

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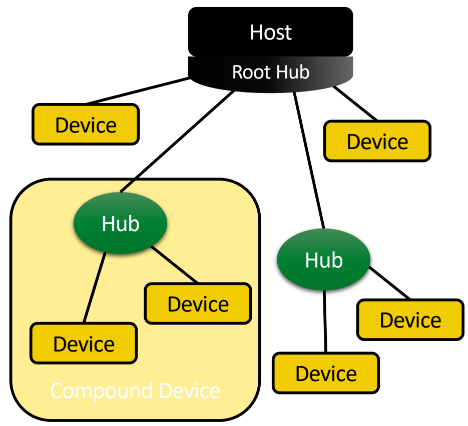
USB – Key Features

- Host-controlled communication model
  - Only one, typically a PC
- Tiered star topology with hubs
- Supports multiple device classes (HID, Mass Storage, Audio, etc.)
- End devices are things like a keyboard, mouse, scanner, web-cam, disk, etc.
- Plug-and-play configuration via descriptors



USB – Device Identification

- Addressing scheme lets the USB host keep track of and communicate with USB devices
- Each USB device is associated with following information:
  - Vendor Identifier (2 bytes)
  - Device Identifier (2 bytes)
  - Version Number (2 bytes)
  - Class Code (3 bytes)
- During initialization, the USB host does the following:
  - Obtains the full address of each USB device
  - Determines required data bandwidth and data types
  - Assigns to each USB a unique 7-bit address

USB Frame Structure

| SYNC | PID | ADDR | ENDP | CRC | DATA | CRC | EOP |

- SYNC: 8-bit (low speed) or 32-bit (high speed) synchronization pattern
- PID: packet ID (token, data, handshake, special)
- ADDR : device address; 7-bits, but `0x00` not used
  - Therefore, can address 127 devices
- ENDP: endpoint, a buffer on a device
  - 4-bits, allowing 16 possible endpoints
- CRC: cyclic redundancy check sum calculated over the data
  - Devices receiving the packet can calculate a local CRC and compare with the transmitted CRC
- EOP: end of packet designator

Token Packet

SYNC	PID	ADDR	ENDP	CRC5	EOP

Data Packet

SYNC	PID	Data 8-1024 B	CRC16	EOP

Handshake Packet

SYNC	PID	EOP

Start of Frame Packet

SYNC	PID	Frame Number	CRC5	EOP

https://www.beyondlogic.org/usbnutshell/usb3.shtml

USB Transfer Modes

- Token-based protocol — Host sends tokens, devices respond
- Transfer modes:
  - **Control**: configuration & commands
    - Initialization of new devices, status queries, etc.
  - **Bulk**: large data (e.g., file transfer)
    - Asynchronous, high jitter and latency possible
  - **Interrupt**: small, time-sensitive data (e.g., mouse)
    - Low latency
  - **Isochronous**: real-time streaming (e.g., audio/video)
    - Low jitter, moderate latency

Next Lecture

- Serial Peripheral Interface – SPI

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Questions

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